



End Semester Assessment SEM II 2025-2026



MSD Internship Review

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SEM II A.Y.(2025-2026)

About MSD

- MSD is a **global biopharmaceutical company** specializing in the research, development, and commercialization of innovative **medicines and vaccines**.
- The company traces its origins to the **late 19th century** and has evolved into one of the **world's leading pharmaceutical manufacturers**, with a strong focus on scientific innovation and public health.
- Its portfolio spans **multiple therapeutic areas**, including **oncology, vaccines, infectious diseases, cardiometabolic disorders, and other serious health conditions**.
- The company is recognized for its longstanding commitment to rigorous research, product quality, patient safety, and ethical business practices.

Product Portfolio

- The company maintains a broad **global footprint**, with commercial operations across **North America, Europe, Asia-Pacific, Latin America, Africa and the Middle East**.
- In the **global oncology market**, MSD's **Keytruda** is a flagship, market-leading product and one of the **top-selling cancer medicines worldwide**, generating approximately **US\$ 31.7 Billion** in annual revenue which is half of the total revenue generated by all products.
- Internally, the **BI solution** consolidates **global and regional Keytruda sales data**, enabling **analysis of revenue trends, KPIs, and market performance over time**.
- **BI solution** provides **multi-level performance visibility**, with my team focusing on **A1-level global business overviews**.

High-Stakes Financial Reporting

- As a flagship of MSD's oncology pipeline, Keytruda generates **massive volumes of financial, sales, and operational data** across diverse global regions.
- The current reporting infrastructure struggles under the weight of this data, causing execution delays that stall time-critical analytical workflows.
- **The Current State & The Bottleneck: 45+ Minute Refresh Latency.**
- The legacy architecture forces Power BI to **execute heavy, complex data transformations (Power Query/DAX)** in addition to rendering visual analytics. This **monolithic approach chokes the compute engine**, creating a severe performance bottleneck that slows down the work of key stakeholders.

The Goal

The Strategic Transformation:

- **Extract & Load (Dataiku):** Ingest raw global oncology financial data securely into Dataiku, shifting the initial staging and integration processes upstream into a robust processing environment.
- **Transform & Clean (Dataiku):** Execute intensive joins, complex aggregations, and business logic outside the BI tool to generate a analysis-ready final table.
- **Visualize & Analyze (Power BI):** Feed the pre-processed, high-performance dataset directly into Power BI.

Timeline

Requirement Analysis

Understand the Global Oncology financial transformation logic, KPIs, filters, and data dependencies.

Planning & Logic Migration

Strategically mapping existing Power Query logic into efficient Dataiku transformation recipes.

QC Checks

Executing rigorous data integrity checks to validate logic parity and ensure absolute financial accuracy.

UAT

(User Acceptance Testing):
Validating the new high-performance dashboards with finance stakeholders for final business sign-off.

Data Migration

Migration of data to the Amazon S3 POC workspace and integration with Dataiku.

Dataiku Pipeline Construction

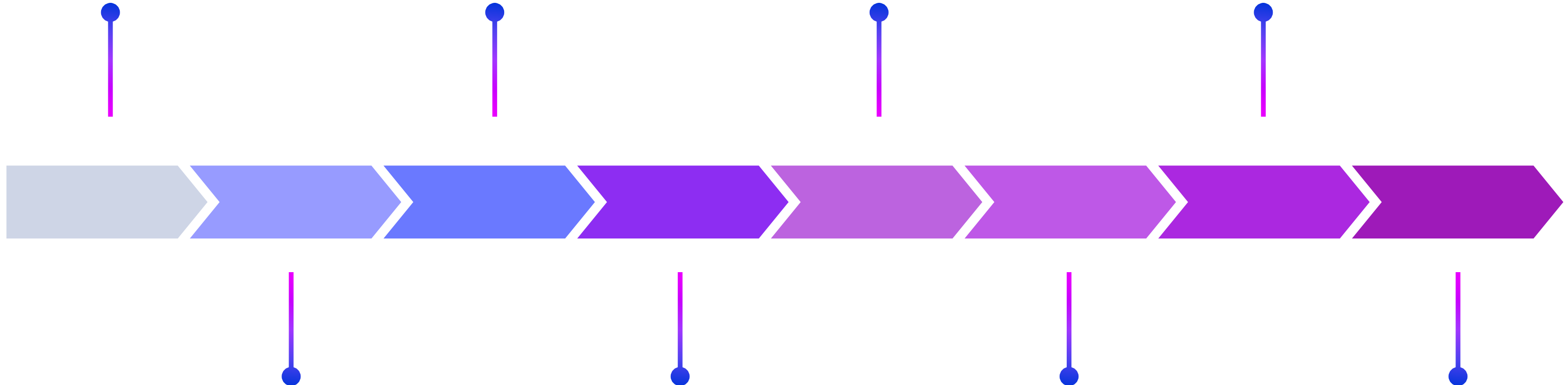
Building the automated ETL flow to execute complex transformations and stage high-quality data.

Power BI Integration

Power BI Integration:
Seamlessly connecting the optimized Dataiku output tables into the existing Power BI reports.

Launch

Deploying the fully optimized, low-latency reporting pipeline into the global production environment.



Tools & Tech Stack



dataiku



pandas



Power Query



Power BI



PySpark

Current Project Status

Completed Milestones:

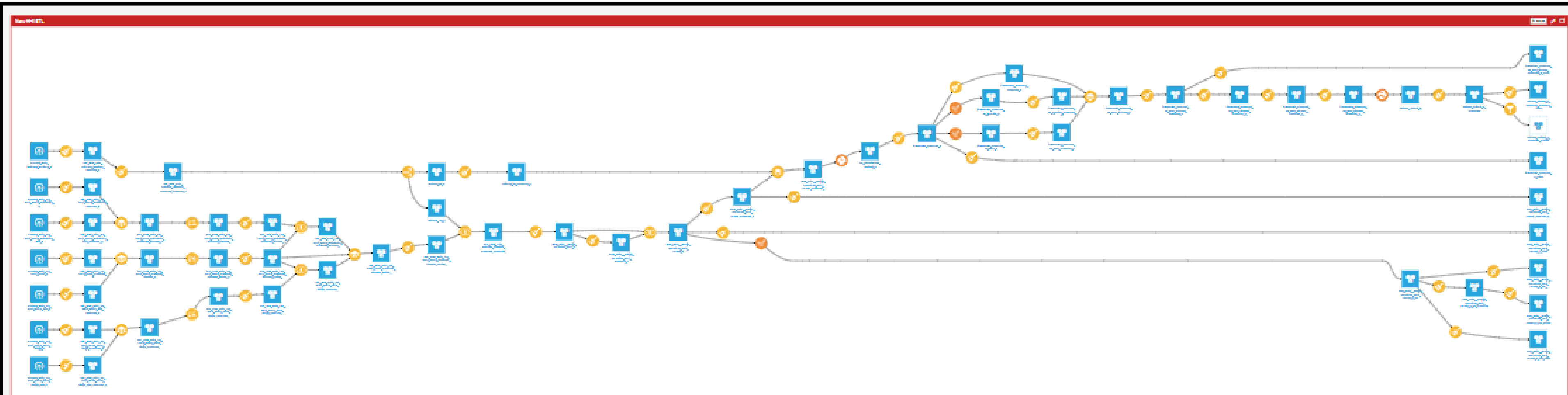
- **Requirement Analysis :** Completed the full architectural analysis of the legacy Power BI Global Oncology Financials report.
- **Data Migration:** Successfully migrated all required raw datasets to the Amazon S3 POC workspace and established Dataiku connectivity.
- **Planning & Logic Migration:** Finished mapping all complex Power Query and DAX transformations into Dataiku recipe logic.
- **Dataiku Pipeline Construction:** The core ETL flow has been built, and primary financial calculations are actively executing within Dataiku.

Current Project Status

Currently In Progress:

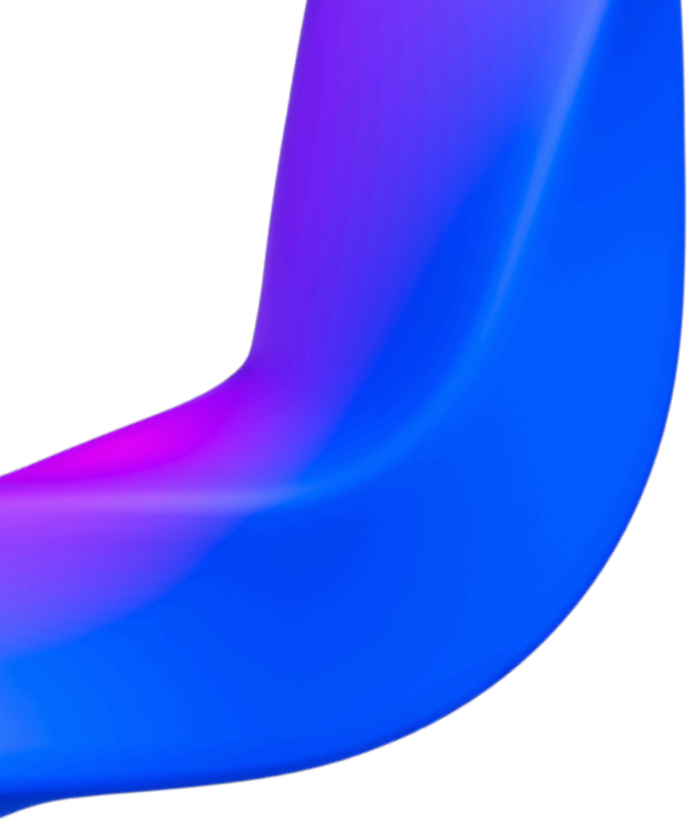
- **Power BI Integration:** The final table from Dataiku is connected into the existing Power BI reports.
- **QC Checks (Active Loop):** We are currently in an iterative Quality Control loop. Initial QC identified specific data discrepancies where new calculations did not perfectly align with legacy outputs.

Dataiku Flow



Summary

- We are transforming Keytruda's global financial reporting **from a high-latency bottleneck into a scalable, high-performance analytical engine.**
- **The Architectural Shift:** By successfully decoupling **heavy ETL workloads** from Power BI and migrating them into Dataiku, we aim to establish a governed, secure data pipeline.
- This optimized architecture is projected to reduce dashboard **refresh times by over 90% (from 45+ minutes to ~5 minutes).**
- **Applying Enterprise Architecture:** Actively transitioning a **monolithic reporting system** into a modern, decoupled data pipeline by separating heavy ETL processing (Dataiku) from visualization (Power BI).



Thank you!